Pre S3 L10-12 Probability Quiz
(ANSWER)

/48

Unit Education

Name: ___________________

Date: ___________________

1. A letter is chosen randomly from the word ‘UNIVERSITY’. Find the probability that is a vowel.

   A. $\frac{1}{5}$

   B. $\frac{3}{10}$

   C. $\frac{2}{5}$

   D. $\frac{1}{2}$

   There are 10 letters in the word ‘UNIVERSITY’.
   There are 4 vowels (U, I, E and I) in the word.
   $\therefore$ Number of favourable outcomes = 4

   $$P(\text{a vowel is chosen}) = \frac{4}{10} = \underline{\underline{\frac{2}{5}}}$$

   C

2. Samuel can choose to go to work by bus, taxi or MTR. Suppose he randomly chooses a mean of transport in two working days. Find the probability that Samuel chooses the same mean of transport in these two working days.

   A. $\frac{2}{3}$

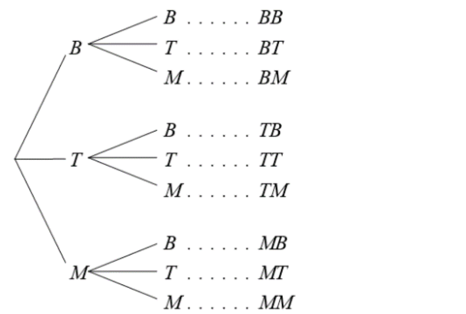
   B. $\frac{1}{3}$

   C. $\frac{1}{4}$

   D. $\frac{3}{5}$

   Let B stand for bus, T stand for taxi and M stand for MTR.

   1st day 2nd day Outcome

   Total number of possible outcomes = 9
   Number of favourable outcomes = 3

   $$\therefore \quad P(\text{same mean of transport}) = \frac{3}{9} = \underline{\frac{1}{3}}$$

   B

3. There are 20 basketball and 35 tennis ball in a box. Nick repeats the action of drawing a ball from the box at random and putting it back into the box. Find the expected number of times of getting a tennis ball if he repeats the action for 385 times.

   A. 20

   B. 35

   C. 140

   D. 245

   $$P(\text{getting a tennis ball}) = \frac{35}{20+35} = \frac{7}{11}$$

   Expected number of times of getting a tennis ball

   $= 385 \times \frac{7}{11}$

   $= \underline{245}$

   D

4. The figure shows a circular fortune wheel in a lucky draw. Dylan turns the wheel once. If the pointer points at the region ‘c’, ’e’ or ’g’, a prize will be given. Find the probability that he gets a prize.

   A. $\frac{1}{4}$

   B. $\frac{1}{6}$

   C. $\frac{1}{10}$

   D. $\frac{1}{12}$

   Let r be the radius of the wheel.

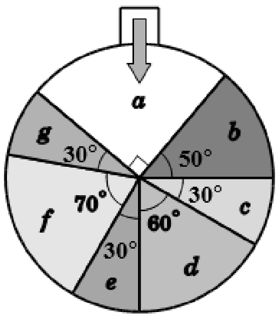
   $P(\text{getting a prize})$

   $$= \frac{\text{sum of the areas of the sectors 'c', 'e' and 'g'}}{\text{area of the wheel}}$$

   $$= \frac{\text{sum of the angles of the sectors 'c', 'e' and 'g'}}{360^\circ}$$

   $$= \frac{30^\circ + 30^\circ + 30^\circ}{360^\circ}$$

   $$= \underline{\frac{1}{4}}$$

   A

5. A bag contains five $\$1.4$ stamps, five $\$0.2$ stamps and ten $\$0.1$ stamps. A stamp is drawn at random from the bag. Find the expected face value of the stamp.

A. $0.35
B. $0.4
C. $0.45
D. $0.5

Total number of possible outcomes = 5 + 5 + 10 = 20

$P(\text{a \$1.4 stamp}) = \frac{5}{20} = \frac{1}{4}$

$P(\text{a \$0.2 stamp}) = \frac{5}{20} = \frac{1}{4}$

$P(\text{a \$0.1 stamp}) = \frac{10}{20} = \frac{1}{2}$

Expected face value of the stamp drawn

$= \$\left(1.4 \times \frac{1}{4} + 0.2 \times \frac{1}{4} + 0.1 \times \frac{1}{2}\right)$

$= \underline{\$0.45}$

C

6. There are 24 bottles of water and x bottles of tea on a table. If a bottle of drink is drawn at random, the probability of drawing a bottle of tea is $\frac{3}{5}$. Find the value of x.

A. 36
B. 38
C. 40
D. 42

$P(\text{a bottle of tea}) = \frac{3}{5}$

i.e. $\frac{x}{x+24} = \frac{3}{5}$

$5x = 3x + 72$

$2x = 72$

$x = \underline{\underline{36}}$

A

7. 6▲ is a 2-digit number, where ▲ is an integer from 0 to 9 inclusive. Find the probability that the 2-digit number is

(a) greater than 60 (3 marks)

(b) divisible by 5 (2 marks)

Total number of possible outcomes = 10 1M

(a) Number of favourable outcomes = 9 1M

$\therefore \quad P(\text{greater than } 60) = \underline{\underline{\frac{9}{10}}}$ 1A

(b) Number of favourable outcomes = 2 1M

$\therefore \quad P(\text{divisible by } 5) = \frac{2}{10}$

$= \underline{\underline{\frac{1}{5}}}$ 1A

8. There are 2000 candidates in an examination. If one of the candidates is chosen randomly, the probability of choosing a female candidate is $\frac{11}{20}$. Find the number of male candidates in the examination. (3 marks)

Let x be the number of female candidates.

$\frac{x}{2\,000} = \frac{11}{20}$ 1M

$x = \frac{11 \times 2\,000}{20}$

$= 1\,100$

$\therefore$ Number of male candidates

$= 2\,000 - 1\,100$ 1M

$= \underline{\underline{900}}$ 1A

9. Two fair dice are thrown at the same time. By using tabulation, find the probability that
   (a) one of the two numbers obtained is a multiple of 3. (5 marks)
   (b) the sum of the two number is less than 9. (2 marks)

The table below lists out all the possible outcomes.

Number on the 2nd dice

Number on the 1st dice	1	2	3	4	5	6
1	(1 , 1)	(1 , 2)	(1 , 3)	(1 , 4)	(1 , 5)	(1 , 6)
2	(2 , 1)	(2 , 2)	(2 , 3)	(2 , 4)	(2 , 5)	(2 , 6)
3	(3 , 1)	(3 , 2)	(3 , 3)	(3 , 4)	(3 , 5)	(3 , 6)
4	(4 , 1)	(4 , 2)	(4 , 3)	(4 , 4)	(4 , 5)	(4 , 6)
5	(5 , 1)	(5 , 2)	(5 , 3)	(5 , 4)	(5 , 5)	(5 , 6)
6	(6 , 1)	(6 , 2)	(6 , 3)	(6 , 4)	(6 , 5)	(6 , 6)

1M+1A

Total number of possible outcomes = $6 \times 6 = 36$ 1M

(a) Number of favourable outcomes = 20 1M

$$\therefore \quad P(\text{one of the numbers obtained is a multiple of 3}) = \frac{20}{36} = \underline{\underline{\frac{5}{9}}}$$ 1A

(b) Number of favourable outcomes = 26 1M

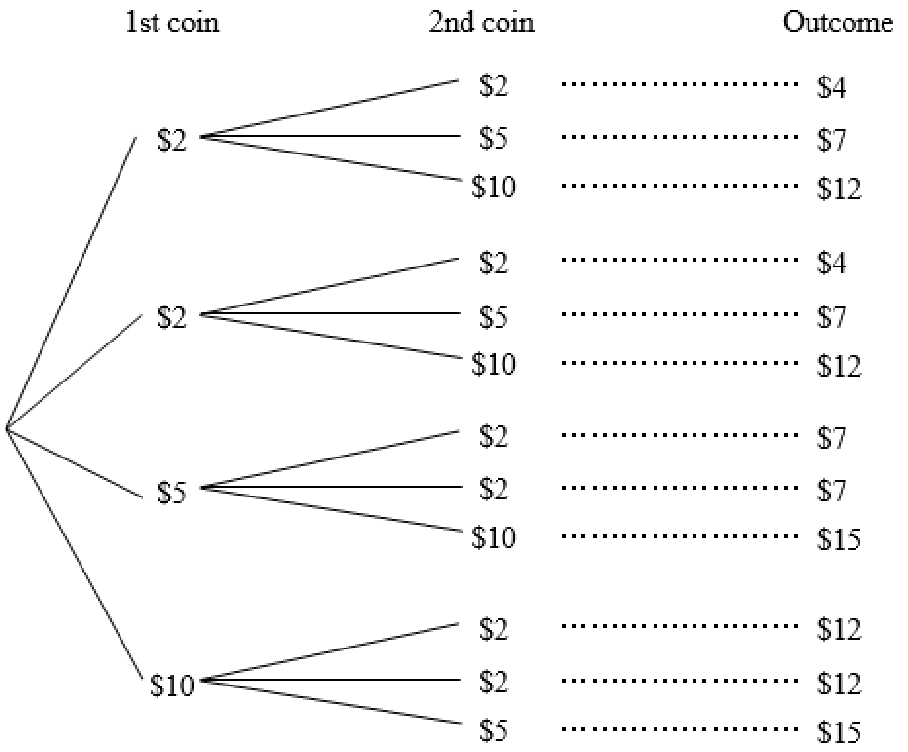
$$\therefore \quad P(\text{the sum is less than 9}) = \frac{26}{36} = \underline{\underline{\frac{13}{18}}}$$ 1A

*10. Winnie's purse contains two $2 coins, one $5 coin and one $10 coin. On a flag day, Winnie takes out two coins randomly from her purse at the same time for donation.

(a) Find the probability that the donation amount is between $5 and $13. (5 marks)

(b) Find the expected donation amount. (4 marks)

(a) The tree diagram below shows all the possible outcomes of the donation amount.



1M+1A

Total number of possible outcomes = 12 1M

Number of favourable outcomes = 8 1M

$\therefore$ P(between $5 and $13) $= \frac{8}{12}$

$= \underline{\frac{2}{3}}$ 1A

(b) $P(\$4) = \frac{2}{12} = \frac{1}{6}$

$P(\$7) = \frac{4}{12} = \frac{1}{3}$

$P(\$12) = \frac{4}{12} = \frac{1}{3}$

$P(\$15) = \frac{2}{12} = \frac{1}{6}$

1M+1A

$\therefore$ Expected donation amount

$= \$\left(4 \times \frac{1}{6} + 7 \times \frac{1}{3} + 12 \times \frac{1}{3} + 15 \times \frac{1}{6}\right)$ 1M

$= \underline{\$9.5}$ 1A